LIVE Environment Inventory

Sanitized project structure and environment-variable reference

This document intentionally excludes live credentials and secret values. Use the accompanying text templates and your approved secret manager to populate local or deployment environments.

Security boundary

The repository contains sensitive values in local-only files such as `.env.local` and `TURN-Server/.env`. Do not copy those values into documents, tickets, chat, source control, or frontend bundles. Rotate any credential that has already been exposed outside its intended secret store.

Project structure

`app/` - Next.js routes for public pages and requester, dispatcher, responder, admin, and auditor workflows.
`components/` - authentication, dashboard, layout, maps, requests, and reusable UI components.
`lib/` - shared types, routing, utilities, mock data/store, and Supabase clients.
`supabase/` - schema and database migrations, including the active-assignment constraint migration.
`TURN-Server/` - optional Coturn Docker deployment and its environment template.
`docs/` - setup and operational documentation.
`scripts/` - local utility scripts, including this PDF generator.

Docker installation and services

Docker item	Details
Service	coturn - optional TURN relay for WebRTC calls
Image	coturn/coturn:latest
Compose file	TURN-Server/docker-compose.yml
Required host software	Docker Desktop or Docker Engine with Docker Compose
Published ports	3478 TCP/UDP; TURN_MIN_PORT-TURN_MAX_PORT UDP (default 49152-49200)
Required env	TURN_USERNAME, TURN_PASSWORD, TURN_REALM, TURN_MIN_PORT, TURN_MAX_PORT

Run from the repository root:

```
Copy-Item TURN-Server/.env.example TURN-Server/.env
docker compose --env-file TURN-Server/.env -f TURN-Server/docker-compose.yml up -d
docker compose --env-file TURN-Server/.env -f TURN-Server/docker-compose.yml ps
docker compose --env-file TURN-Server/.env -f TURN-Server/docker-compose.yml down
```

The current Docker configuration installs and runs Coturn only. It does not containerize the Next.js app or Supabase; those are configured separately in this repository. Replace the image tag with a pinned version for production after testing.

What the TURN server does

WebRTC tries to connect two callers directly. If NAT, a firewall, or different networks prevent a direct connection, the TURN server relays the audio and video traffic between them. Coturn is the relay service used here. It is a fallback path, not a replacement for the application, Supabase, or the browser WebRTC code.

To get it working: (1) run Docker on a reachable host, (2) set a strong TURN username and password plus a reachable TURN realm or hostname, (3) allow TCP/UDP 3478 and UDP 49152-49200 through the host firewall and router, (4) start the Compose service, and (5) add the TURN URL and credentials to the app's WebRTC iceServers configuration. The current repository contains the server setup, but the app is not yet connected to TURN automatically, so step 5 is still required before calls will use it.

For local testing on one computer or the same Wi-Fi network, TURN may not be necessary. For production, use a public DNS name or IP, TLS where appropriate, and short-lived TURN credentials instead of a permanent frontend password.

Install and start the TURN server

Step 1 - Install Docker Desktop. On Windows, download it from https://www.docker.com/products/docker-desktop/ and run the installer with the default settings. Restart Windows if requested, open Docker Desktop, and wait until it reports that Docker is running. Then verify the installation in PowerShell:

```
docker --version
docker compose version
```

Step 2 - Get the project. Clone the repository or open the folder downloaded from GitHub, then open PowerShell in the project root. The project root is the folder containing `package.json` and `TURN-Server`.

Step 3 - Create the TURN environment file and fill in the values from your secret manager:

```
Copy-Item TURN-Server/.env.example TURN-Server/.env
notepad TURN-Server/.env
```

Set TURN_USERNAME, TURN_PASSWORD, TURN_REALM, TURN_MIN_PORT, and TURN_MAX_PORT. Do not commit this file or paste its secrets into the PDF.

Step 4 - Allow the network ports. On the TURN host and router, allow 3478 TCP and UDP plus the UDP relay range configured by TURN_MIN_PORT and TURN_MAX_PORT. For the included example, that range is 49152-49200. A public DNS name or reachable public IP is required for callers on other networks.

Step 5 - Start Coturn and check it:

```
docker compose --env-file TURN-Server/.env -f TURN-Server/docker-compose.yml up -d
docker compose --env-file TURN-Server/.env -f TURN-Server/docker-compose.yml ps
docker compose --env-file TURN-Server/.env -f TURN-Server/docker-compose.yml logs -f coturn
```

Step 6 - Connect the app. Add the TURN server URL, username, and short-lived credential to the application's WebRTC `iceServers` list. Restart the Next.js app after changing environment variables, then test a call from two different networks. If the container is running but calls do not relay, check the public hostname, firewall/router forwarding, and the relay port range.

Main project environment

Variable	Scope / purpose	Value in this document
NEXT_PUBLIC_SUPABASE_URL	Browser + server Supabase project URL	[REDACTED]
NEXT_PUBLIC_SUPABASE_ANON_KEY	Browser Supabase public key	[REDACTED]
SUPABASE_SERVICE_ROLE_KEY	Server-only Supabase privileged key	[REDACTED]
JWT_SECRET / JWT_ISSUER / JWT_AUDIENCE	Server token signing and validation	[REDACTED]
ACCESS_TOKEN_TTL_SECONDS / REFRESH_TOKEN_TTL_SECONDS	Token lifetimes	[REDACTED]
NODE_ENV / LOG_LEVEL / ALLOW_INSECURE_COOKIES	Runtime behavior and logging	[REDACTED]
ROUTING_MAX_RADIUS_KM / RATE_LIMIT_ENABLED	Routing and rate-limit controls	[REDACTED]
NEXT_PUBLIC_MAP_* / NEXT_PUBLIC_ROUTE_LINK_PROVIDER	Browser map and route configuration	[REDACTED]
NEXT_PUBLIC_TRACKING_* / NEXT_PUBLIC_POLL_INTERVAL_MS	Browser tracking intervals	[REDACTED]
GEMINI_API_KEY / GEMINI_MODEL	Optional requester AI	[REDACTED]
OLLAMA_BASE_URL / OLLAMA_MODEL	Optional local AI	[REDACTED]
GMAIL_USER / GMAIL_APP_PASSWORD	Mail / notification integration	[REDACTED]
NEXT_PUBLIC_TURN_URL / NEXT_PUBLIC_TURN_USERNAME / NEXT_PUBLIC_TURN_CREDENTIAL	Optional WebRTC TURN client settings	[REDACTED]

TURN server environment

Variable	Used by	Value
TURN_USERNAME	Coturn long-term credential username	[REDACTED]
TURN_PASSWORD	Coturn long-term credential password	[REDACTED]
TURN_REALM	Coturn realm / reachable host identity	[REDACTED]
TURN_MIN_PORT / TURN_MAX_PORT	UDP relay port range	[REDACTED]

`TURN-Server/docker-compose.yml` passes these values to Coturn and publishes 3478 TCP/UDP plus the configured relay UDP range. Keep the TURN password server-side and prefer short-lived credentials for production.

Operational notes

1. Copy `docs/env.project.template.txt` to the appropriate local/deployment environment file and populate it from the secret manager.
2. Copy `docs/env.turn-server.template.txt` to `TURN-Server/.env` only on the TURN host.
3. Confirm secret files are ignored by Git before committing.
4. Never place server-only keys in variables prefixed `NEXT_PUBLIC_`.
5. If credentials were shared accidentally, revoke and rotate them at the provider before continuing.

Generated companion files

The repository now includes sanitized, copy-ready templates for the main project and TURN server. They preserve variable names and deployment intent but contain placeholders instead of live credentials.